
UNIT 9 EXCEL FOR DESCRIPTIVE DATA ANALYSIS

Structure

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9.0 INTRODUCTION

In the world of data analysis, the ability to summarize and interpret data effectively is crucial. Descriptive data analysis serves as the foundation for understanding datasets, allowing analysts to uncover patterns, trends, and insights that may not be immediately obvious. Microsoft Excel, one of the most widely used tools for data analysis, provides a versatile platform for performing descriptive statistics.

This unit will guide you through the essentials of using Excel for descriptive data analysis, covering various tools and functions that enable you to explore and summarize data efficiently. From calculating basic summary statistics such as mean, median, and standard deviation to creating visualizations like histograms and box plots, Excel offers an intuitive interface to analyze and present data.

Whether you're a beginner looking to get acquainted with Excel's analytical capabilities or an experienced user seeking to refine your skills, this chapter will equip you with the knowledge needed to perform a comprehensive descriptive analysis of any dataset. You will learn how to leverage Excel's built-in functions, data tools, and visualization options to transform raw data into meaningful insights that can guide informed decision-making.

By the end of this chapter, you will be able to confidently navigate Excel's data analysis features and use them to perform essential descriptive statistics that are critical in data-driven environments.

9.1 OBJECTIVES

After finishing this unit, you will be able to:

- Describe the meaning

9.2 IMPORTING DATA FROM VARIOUS FILE FORMATS

In the field of data analysis, the ability to seamlessly import data from a variety of file formats is an essential skill. Data often comes in different forms, such as spreadsheets, text files, databases, or even web-based sources. Each of these file types presents unique challenges and opportunities when it comes to importing data into Excel for analysis.

This section will focus on how to import data from various commonly used file formats into Excel. Whether you are working with CSV, TXT, or database files like Access or SQL, Excel provides a range of options to facilitate smooth data importation. Understanding how to properly bring external data into Excel is crucial for ensuring the integrity and accuracy of your analysis.

We will explore the step-by-step processes for importing data from different formats, discussing best practices, potential pitfalls, and tips for maintaining data consistency and quality throughout the import process. By mastering this skill, you'll be able to work with diverse datasets from a variety of sources, making Excel a powerful tool for your data analysis tasks.

Here's how you can import data from different file formats into Excel:

1. Importing from a Text or CSV File

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **From Text/CSV** in the **Get & Transform Data** group.
- c) **Browse** and select your file, then click **Import**.
- d) **Preview the data** and make any necessary changes in the Import Wizard.
- e) Click **Load** to import the data into Excel.

2. Importing from an XML File

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From File**, and select **From XML**.
- c) **Browse** and select your XML file, then click **Import**.
- d) **Select the elements** you want to import and click **OK**.
- e) Click **Load** to import the data into Excel.

3. Importing from an Access Database

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From Database**, and select **From Microsoft Access Database**.
- c) **Browse** and select your Access database file, then click **Import**.
- d) **Select the tables** you want to import and click **Load**.

4. Importing from a Web Page

- a) **Open Excel** and go to the **Data** tab.

- b) Click on **Get Data**, then **From Other Sources**, and select **From Web**.
- c) **Enter the URL** of the web page you want to import data from and click **OK**.
- d) **Select the data** you want to import from the Navigator window.
- e) Click **Load** to import the data into Excel.

5. Importing from JSON

- a) **Open Excel** and go to the **Data** tab.
- b) Click on **Get Data**, then **From File**, and select **From JSON**.
- c) **Browse** and select your JSON file, then click **Import**.
- d) **Select the data** you want to import from the Navigator window.
- e) Click **Load** to import the data into Excel.

These steps should cover most common file formats you might want to import into Excel.

Check Your Progress 1

- 1. What file types can Excel import data from?
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- 2. What is the importance of maintaining data consistency during import?
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- 3. What steps are followed to import data from a CSV file?
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- 4. How can data be imported from a web page into Exce?
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9.3 DATA HANDLING AND PRE-PROCESSING

Data handling and pre-processing are essential steps in the data analysis pipeline, though they refer to different aspects of preparing data for analysis. While both involve transforming raw data into a clean, structured form, each process has a distinct focus.

Data Handling refers to the management and organization of data within a dataset. It involves tasks such as sorting, filtering, and structuring data to make it easier to analyze. For instance, data handling can include organizing columns, merging multiple datasets, or converting data into a more useful format. The goal is to ensure that the dataset is structured logically and efficiently, often by removing or consolidating irrelevant or redundant information.

On the other hand, **Data Pre-Processing** focuses more on cleaning the dataset to ensure its quality and reliability. This includes handling missing values, correcting errors, dealing with outliers, and ensuring consistency across the data. Pre-processing may also involve transforming variables or normalizing data to make it comparable across different units of measurement. The objective is to remove any issues that could distort or skew analysis results, ultimately ensuring the data is accurate and suitable for the intended analysis.

While both data handling and pre-processing are critical to the success of any data analysis project, the main difference lies in their focus. Data handling is primarily concerned with the organization and structuring of data, whereas data pre-processing focuses on cleaning and preparing the data to ensure its quality. In practice, the two processes often overlap, with one leading naturally into the other. For example, handling missing values (a pre-processing task) often involves reorganizing or filtering the dataset (a data handling task).

Before any meaningful analysis can be performed, the raw data must be properly prepared and cleaned. Data handling and pre-processing are critical steps in ensuring that the information you are working with is accurate, consistent, and ready for analysis. In Excel, this process involves organizing, transforming, and cleaning data to eliminate errors, deal with missing values, and ensure that the dataset is structured correctly.

This section will guide you through the essential techniques for handling and pre-processing data within Excel. We will cover methods for cleaning datasets, such as removing duplicates, handling missing values, and correcting inconsistencies. Additionally, you will learn how to reshape data, filter out irrelevant information, and apply transformations that make the data more useful for analysis.

Effective data handling and pre-processing not only improve the quality of your analysis but also save time and effort in the long run. By the end of this section, you will be equipped with the tools and strategies needed to prepare your data for further analysis, ensuring that your findings are built on a solid foundation of clean, reliable information.

In short, both data handling and pre-processing are integral to preparing data for analysis. Properly handling and cleaning data in these early stages helps analysts avoid pitfalls that could lead to inaccurate conclusions, ultimately ensuring more reliable and insightful results from the analysis.

Data Handling: Data handling refers to the broader process of managing data throughout its lifecycle. It includes tasks like:

- **Data Collection:** Gathering data from various sources.
- **Data Storage:** Organizing and storing data in databases, spreadsheets, or other storage systems.
- **Data Access:** Retrieving and making data available for analysis or other purposes.
- **Data Management:** Ensuring data quality, consistency, and security.
- **Data Sharing:** Distributing data to stakeholders as needed.

Data Preprocessing: Data preprocessing is a subset of data handling focused specifically on preparing raw data for analysis. It involves tasks like:

- **Data Cleaning:** Removing duplicates, correcting errors, handling missing values, and normalizing data.
- **Data Transformation:** Converting data into a suitable format, such as aggregating, encoding, or scaling data.
- **Data Integration:** Combining data from different sources to create a unified dataset.
- **Data Reduction:** Reducing the volume of data by selecting relevant features or using dimensionality reduction techniques.

Key Differences between Data Handling and Data Preprocessing are:

- **Scope:** Data handling is broader and encompasses the entire lifecycle of data, while data preprocessing is specifically about preparing data for analysis.
- **Focus:** Data handling includes management, storage, and sharing aspects, whereas preprocessing focuses on cleaning and transforming data for analytical purposes.
- **Stage:** Data handling happens at all stages of the data lifecycle, while preprocessing is typically an initial step before data analysis.

Some frequently used Excel functions, commands and formulas for Data Handling: there is a fundamental difference between the functions, commands and formulas for Data Handling, to begin with, firstly understand the difference among them.

1. **Excel Functions for Data Handling : Functions** in Excel are predefined formulas that perform specific operations on data. They are built-in operations that can be directly used in cells to carry out calculations, manipulate text, or analyze data. Functions are used within formulas to perform calculations, making them a fundamental part of data handling.

For example, **TEXT functions** such as =TRIM(A1) and =LEFT(A1, 3) can clean or extract parts of text data. Statistical functions like =SUM(A1:A10) and =AVERAGE(B1:B10) help you quickly summarize data. Lookup functions, such as =VLOOKUP(A1, B1:C10, 2, FALSE), allow you to search for data within a range. Logical functions, such as =IF(A1 > 10, "Yes", "No"), help you evaluate conditions in your data.

Excel functions are essential because they allow you to manipulate and retrieve data directly inside cells without needing to manually process it. They are predefined operations that return a specific value based on the arguments provided.

2. **Excel Commands for Data Handling: Commands** are actions or tools within the Excel interface that enable users to perform data-related tasks through the ribbon, context menu, or keyboard shortcuts. Unlike functions, commands don't require entering formulas in cells but are instead applied to the dataset via Excel's built-in features.

For example, **Sort** is a command that organizes data in ascending or descending order, which can be accessed through the "Data" tab. The **Filter** command helps display only rows that meet certain criteria, also found in the "Data" tab. The **Remove Duplicates** command eliminates duplicate entries in a dataset, ensuring that only unique values remain. **Text to Columns** is another command that splits data from one column into multiple columns based on a delimiter, such as commas or spaces.

Commands are used to alter the structure of data or modify how it is displayed, often without needing to write any formulas. They provide quick ways to reorganize, clean, or modify data directly through the Excel interface.

3. **Excel Formulas for Data Handling: Formulas** in Excel combine one or more functions, cell references, and operators to perform more complex data manipulations or calculations. A formula is an expression entered into a cell, which can be as simple as adding two values or as complex as combining multiple functions for advanced data analysis.

For example, a basic formula such as `=A1 + B1` adds the values in two cells. More complex formulas might combine functions, such as `=IF(A1 > 10, "Pass", "Fail")`, which evaluates a condition and returns one result if true and another if false. Nested formulas, like `=SUM (IF (A1:A10 > 5, B1:B10))`, allow for multiple levels of conditions. **Array formulas**, such as `{=SUM (A1:A10 * B1:B10)}`, handle multiple values simultaneously and perform operations on entire arrays.

Formulas provide the flexibility to customize data handling, allowing you to create specific calculations or manipulations by combining various functions and operators. They are dynamic and can be adapted to a wide range of tasks, depending on the needs of the data analysis.

Key Differences between Function, Command and Formula:

- **Functions** are predefined tools in Excel that are used within formulas to perform specific operations on data. They are written inside cells and return a result based on their input.
- **Commands** are actions performed through the Excel interface (via ribbon, menus, or shortcuts). They modify data directly but don't require typing formulas in cells.
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- **Formulas** are combinations of functions, cell references, and operators. They allow users to perform complex calculations and data manipulations by combining multiple elements into a single expression.

Each of these tools—**functions**, **commands**, and **formulas**—play a crucial role in data handling within Excel, making it easier to organize, analyze, and manipulate datasets according to specific needs.

Frequently Used Excel Functions for Data Handling & Pre-Processing

These are built-in formulas that help with organizing and manipulating data directly in the cells:

1. **TEXT Functions** (for manipulating and cleaning text data):
 - **LEFT**: Extracts a specified number of characters from the beginning of a text string.
 - Example: =LEFT(A1, 5)
 - **RIGHT**: Extracts characters from the end of a text string.
 - Example: =RIGHT(A1, 3)
 - **MID**: Extracts a substring from the middle of a text string.
 - Example: =MID(A1, 3, 4)
 - **TRIM**: Removes extra spaces from a text string.
 - Example: =TRIM(A1)
 - **UPPER**: Converts text to uppercase.
 - Example: =UPPER(A1)
 - **LOWER**: Converts text to lowercase.
 - Example: =LOWER(A1)
 - **PROPER**: Capitalizes the first letter of each word.
 - Example: =PROPER(A1)
 - **TEXTJOIN**: Joins text from multiple cells using a specified delimiter.
 - Example: =TEXTJOIN(" ", TRUE, A1:A5)
2. **Lookup and Reference Functions** (for retrieving and referencing data):
 - **VLOOKUP**: Searches for a value in the first column and returns a value in the same row from another column.
 - Example: =VLOOKUP(A2, B2:D10, 3, FALSE)
 - **HLOOKUP**: Searches for a value in the first row and returns a value in the same column from another row.
 - Example: =HLOOKUP(A1, B1:F3, 2, FALSE)
 - **INDEX**: Returns the value of a cell in a specified row and column from a given range.
 - Example: =INDEX(A1:C10, 2, 3)

- **MATCH**: Searches for a value in a range and returns the relative position.
 - Example: =MATCH("John", A2:A10, 0)
- **OFFSET**: Returns a cell or range of cells that is a specified number of rows and columns from a given reference.
 - Example: =OFFSET(A1, 2, 3)

3. Logical Functions (for conditional handling):

- **IF**: Performs a logical test and returns one value if true and another if false.
 - Example: =IF(A1 > 50, "Pass", "Fail")
- **COUNTIF**: Counts the number of cells that meet a specific condition.
 - Example: =COUNTIF(A1:A10, ">5")
- **SUMIF**: Sums the values in a range based on a specific condition.
 - Example: =SUMIF(A1:A10, ">5", B1:B10)
- **AND/OR**: Returns TRUE if all/any of the conditions are true.
 - Example: =AND(A1 > 50, B1 < 100)

4. Aggregation Functions (for summarizing data):

- **SUM**: Adds up all the values in a range.
 - Example: =SUM(A1:A10)
- **AVERAGE**: Calculates the average of a range.
 - Example: =AVERAGE(A1:A10)
- **MAX/MIN**: Returns the highest or lowest value in a range.
 - Example: =MAX(A1:A10) and =MIN(A1:A10)
- **MEDIAN**: Returns the median value of a range.
 - Example: =MEDIAN(A1:A10)
- **SUBTOTAL**: Returns a subtotal for a range, allowing different aggregation functions (SUM, AVERAGE, etc.) depending on the function number.
 - Example: =SUBTOTAL(9, A1:A10) (9 represents the SUM function)

5. Unique and Distinct Data Functions:

- **UNIQUE** (Excel 365 and Excel 2021): Returns a list of unique values from a range.
 - Example: =UNIQUE(A1:A10)
- **REMOVE DUPLICATES**: Removes duplicate values from a selected range (Command, not a function).

Frequently Used Excel Commands for Data Handling & Pre-Processing

These are features you can access via Excel's ribbon, context menus, or keyboard shortcuts for organizing and manipulating data:

1. **Sort:** Organizes data in ascending or descending order based on one or more columns.
 - How to use: Go to the "Data" tab and select "Sort."
2. **Filter:** Displays only the rows that meet specific criteria.
 - How to use: Go to the "Data" tab and click "Filter."
3. **Remove Duplicates:** Eliminates duplicate entries in a dataset, ensuring each record is unique.
 - How to use: Go to the "Data" tab and click "Remove Duplicates."
4. **Text to Columns:** Splits data from one column into multiple columns based on delimiters (such as commas or spaces).
 - How to use: Go to the "Data" tab and select "Text to Columns."
5. **Flash Fill:** Automatically fills in data based on patterns detected from your initial entries.
 - How to use: Start typing a pattern and press Ctrl + E to trigger Flash Fill.
6. **Group and Outline:** Organizes data into groups that can be expanded or collapsed.
 - How to use: Select rows/columns, right-click, and choose "Group."
7. **Remove Blank Cells:** Select and remove blank cells in a range.
 - How to use: Use the "Go To Special" command (F5 or Ctrl + G), select "Blanks," and delete or replace them.
8. **Consolidate:** Combines data from multiple ranges into one summary report.
 - How to use: Go to the "Data" tab and select "Consolidate."
9. **Data Validation:** Restricts the type of data that can be entered into a cell.
 - How to use: Go to the "Data" tab and click on "Data Validation."
10. **PivotTables:** Summarizes and analyzes large datasets by dynamically grouping and aggregating data.
 - How to use: Go to the "Insert" tab and click on "PivotTable."
11. **Hyperlink:** Adds a clickable link to a specific URL or location within a workbook.
 - How to use: Go to the "Insert" tab and select "Hyperlink."

Frequently Used Excel Formulas for Data Handling & Pre-Processing

These formulas are used to combine functions and logic to handle complex data tasks.

1. **Concatenate (or CONCAT):** Combines the contents of two or more cells into one.
 - Formula: =CONCAT(A1, " ", B1) or =CONCATENATE(A1, " ", B1)

2. **IFERROR:** Returns a specified value if a formula results in an error.
 - Formula: =IFERROR(VLOOKUP(A2, B2:B10, 1, FALSE), "Not Found")
3. **INDEX and MATCH Combination:** An alternative to VLOOKUP for more flexible data referencing.
 - Formula: =INDEX(B2:B10, MATCH(A2, A1:A10, 0))
4. **SUMPRODUCT:** Multiplies corresponding values in ranges or arrays and returns the sum of the products.
 - Formula: =SUMPRODUCT(A1:A10, B1:B10)
5. **ARRAY Formulas:** Allows for complex calculations on multiple values at once.
 - Formula: {=SUM(A1:A10 * B1:B10)} (entered using Ctrl + Shift + Enter)

The data pre-processing also involves various tasks including data cleaning, transformation, filtering, formatting, integration, aggregation, and visualization also. The excel possesses function, formulas and commands for each, bellow is a brief description for the key pre-processing tasks you can perform in excel:

1. Data Cleaning:

- **Removing Duplicates:** Use the **Remove Duplicates** feature in the Data tab.
- **Handling Missing Values:** Replace, fill, or remove missing values using functions like **IFERROR**, **ISBLANK**, and **IFNA**.
- **Data Validation:** Create rules to ensure data integrity by using the **Data Validation** feature.

2. Data Transformation

- **Text Functions:** Use functions like **CONCATENATE**, **LEFT**, **RIGHT**, **MID**, and **TEXT** for text manipulation.
- **Date and Time Functions:** Use functions like **DATEDIF**, **DATE**, **YEAR**, **MONTH**, and **DAY** for handling dates and times.
- **Mathematical Functions:** Apply functions such as **SUM**, **AVERAGE**, **ROUND**, **FLOOR**, and **CEILING** for numerical data processing.

3. Data Filtering and Sorting

- **Filters:** Use the **Filter** feature to narrow down data based on specific criteria.
- **Sorting:** Organize data in ascending or descending order based on one or more columns.

4. Data Formatting

- **Conditional Formatting:** Highlight cells that meet specific criteria using color scales, data bars, and icon sets.
- **Cell Formatting:** Adjust cell formats to enhance data readability,

such as changing number formats, font styles, and alignment.

5. Data Integration

- **Merging Data:** Use **VLOOKUP**, **HLOOKUP**, **INDEX**, and **MATCH** to combine data from different sources.
- **Appending Data:** Copy and paste data from multiple worksheets into a single worksheet.

6. Data Aggregation

- **Pivot Tables:** Summarize large datasets and generate insights using **Pivot Tables**.
- **Grouping:** Group data by categories or time periods to make analysis easier.

7. Data Visualization

- **Charts and Graphs:** Create various types of charts (like bar, line, pie) to visualize data trends and patterns.
- **Slicers and Timelines:** Use slicers and timelines to filter data dynamically in Pivot Tables and Pivot Charts.

8. Power Query: For more advanced preprocessing, use **Power Query**:

- **Data Import and Transformation:** Import data from various sources and perform complex transformations.
- **Advanced Cleaning:** Automate data cleaning tasks like removing duplicates, unpivoting data, and splitting columns.
- **Data Shaping:** Reshape and reorganize data to fit your analysis needs.

Excel's robust set of features makes it a powerful tool for data preprocessing. Whether you're dealing with small datasets or large ones, it has the tools you need to clean, transform, and prepare your data for analysis.

☞ Check Your Progress 2

1. How do data handling and pre-processing differ?

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2. What are common Excel functions for handling text data?

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3. What is the role of commands like 'Remove Duplicates' and 'Sort'?

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4. What Excel features help with missing or incorrect data?

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9.4 FREQUENCY DISTRIBUTION

A **frequency distribution** is a summary of how often different values or ranges of values occur in a dataset. It helps in understanding the distribution and patterns in data, such as how frequently each value or range appears.

To understand the process of generation of frequency distribution in excel consider a dataset given in the Example-1 below:

Example-1: Below, Table 1 - captures the lives of 100 electric bulbs (in Hrs.), and say, you are required to construct a frequency distribution and create histogram from this data using Spreadsheet package.

Table 1: Lives of 100 Electric Bulbs

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

Solution: Now to summarize the pattern in dataset we need to develop a **Frequency Distribution, which** is the most common method for summarizing and presentation of data, which enables us to quickly assess how frequently any value occurs in the given data set.

Figure 1 shows the spreadsheet (for the example, MS-Excel has been used, but you are free to use any spreadsheet package). Enter the given data in the cells $A2....E21$, that needs to be used to construct the frequency distribution. In the next step, you are required to specify class intervals to construct a frequency distribution. In this context, observe that the minimum and maximum values for the raw data are 511.6 and 1314.7 respectively (use $\min(A2:E21)$, $\max(A2:E21)$), as these are useful information in deciding about the classes to be formed. Here, the data under bins shows the class categories and they can be read as 0 to 510.5 (included), 510.6 to 590.5 (included) and so on. The last category is 1390.6 and above. Please note that for the purpose of calculation of frequencies, the upper limits of the classes ($H2....H13$) alone are required.

Once the basic data is entered, you need to calculate the Frequency distribution for these class intervals. For this, you need to enter an array formula in the spreadsheet. An array formula performs calculations on an array of data and may produce an array of results. To enter array formula, you may need to press CTRL+SHIFT+ENTER into a worksheet. The formula for calculating frequency is also an array formula. To enter this formula, perform the following steps:

- Select the cells in which result is expected in our case these are $J2....J14$.
- Enter the formula: $=\text{frequency}(A2:E21,H2:H13)$ and press CTRL+SHIFT+ENTER keys.
- You can also find the Cumulative frequencies, use simple addition formula for this purpose in cells $K2....K14$. Write your own formula for this purpose.

On completion of these operations, the screen will look like as that of Figure 1.

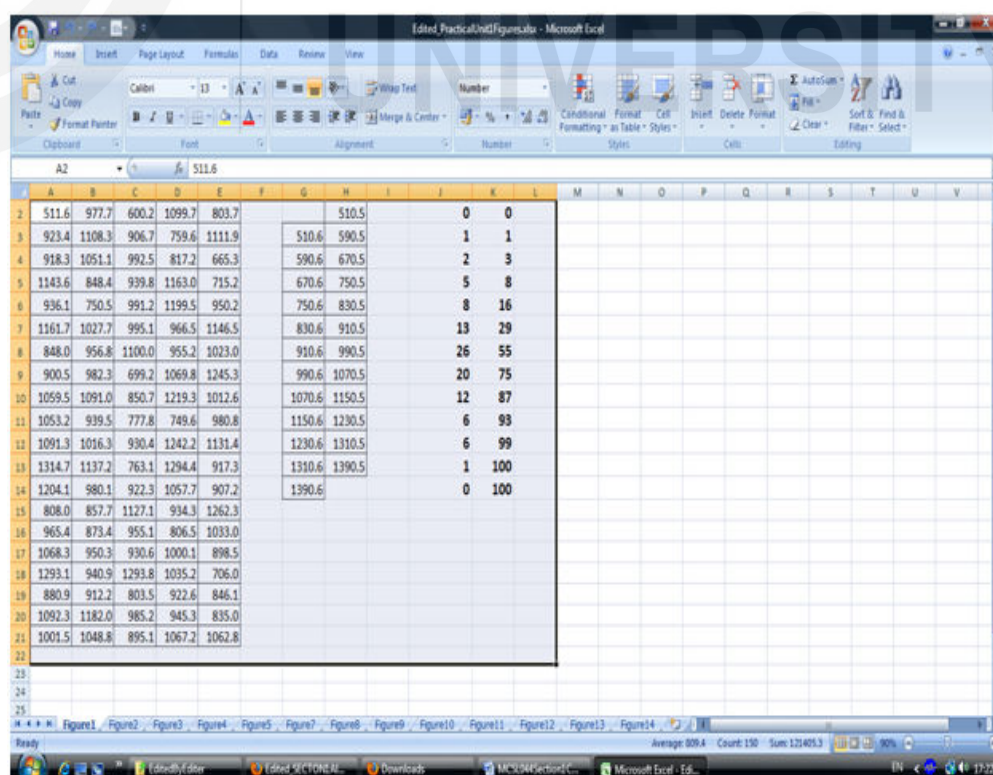


Figure 1: Calculating Frequencies in Spreadsheet

Now, to create a histogram as well as frequency distribution, you may also use *Data → Data Analysis* Menu. In some of your worksheets this option may not be displayed. In such case follow the following steps to *Load the Analysis ToolPak*.

Steps to Activate Data Analysis Toolpak

For Windows (Excel 2016, 2019, 2021, Microsoft 365):

1. Open Excel.
2. Go to **File > Options**.
3. In the **Excel Options** dialog box, select **Add-ins**.
4. At the bottom, next to **Manage**, select **Excel Add-ins** from the dropdown, then click **Go**.
5. In the **Add-Ins** box, check **Analysis ToolPak**.
6. Click **OK**.

Once activated, you'll see "**Data Analysis**" on the **Data** tab (usually on the far right). On loading this, you will be able to see the *Data Analysis* option in the *Data* Menu option.

In order to make the histogram, you just need to use the original raw data and bins, so delete all other data in Figure 1 except keep the columns *A* to *E* and *H*. Moreover, transfer the column *H* data to *G* by incrementing previous *G* data by 0.1. Now to make the histogram perform the following steps:

1. Select *Data → Data Analysis* and then select *Histogram* in the resulting dialog box and press **Ok** button.
2. In the resulting dialog box, set the *Input Range* to *A2..E21*, *Bin Range* to *G2..G13* and *Output Range* to *I1..J14*. Please do not forget to check the *Chart Output* check box.
3. Click **Ok**

The Histogram will be displayed as shown in the Figure 2. You need to modify the format of the histogram to make it look as per your need. For example, to remove gaps in between bars:

1. Click on Bar → *Right Click → Format Data Series...*
2. *Series Options → Gap Width 0%*
3. *Fill → Pattern Fill*
4. *Border Colour → Solid Line etc..*

Notice that horizontal axis labels now depict the upper class boundaries having class width of 80 hours in Figure 2. Can you now plot the frequency polygon and the less than type to give for the same frequency distribution You can also plot Bar diagram for appropriate data as shown in the Unit.

You can also explore the fact that all spreadsheet packages allow you to modify the Title of the chart, Axis titles, colour of bars, size of bars, size of chart and many other formatting features. For this you should refer to documentation of the spreadsheet package that you are using.

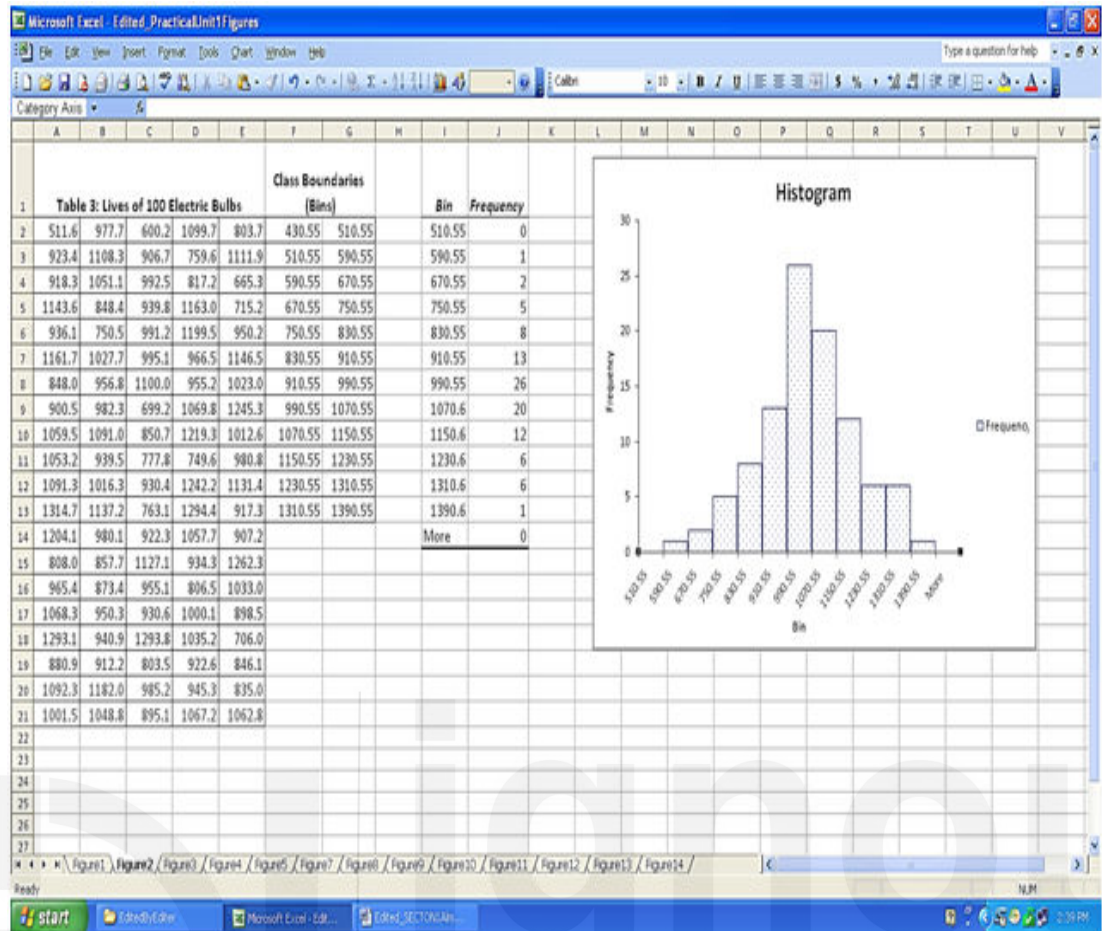


Figure 9.2: Histogram using the Data → Data Analysis options

☞ Check Your Progress 3

1. What is the purpose of a frequency distribution?

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2. How is the FREQUENCY function used in Excel?

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3. What tool is used in Excel to create histograms?

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4. How do you determine class intervals for a frequency distribution?

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9.5 CENTRAL TENDENCY AND DISPERSION

Both Central tendency and dispersion are the fundamental concepts in statistics used to summarize and describe datasets.

Central tendency refers to the measure that identifies the center or typical value within a dataset. It provides a single value that attempts to describe a set of data by indicating the position around which most values cluster. The most common measures of central tendency are the mean, which is the arithmetic average of all values; the median, which is the middle value when the data is arranged in order; and the mode, which is the value that appears most frequently. These measures give a general idea of where the data points tend to concentrate.

To find the mean, which is the average of a dataset, you can use the formula `=AVERAGE(range)`, where "range" refers to the set of cells containing your data. For example, `=AVERAGE(A2:A10)` will return the average value of the numbers in cells A2 through A10.

To determine the median, or the middle value when your data is sorted, use `=MEDIAN(range)`. This function is especially useful when your data has outliers that may skew the mean.

To find the mode, which is the most frequently occurring number in a dataset, use `=MODE.SNGL(range)`. If your dataset has more than one mode, you can use `=MODE.MULT(range)` to return multiple results.

On the other hand, dispersion refers to the degree of spread or variability in the data. While central tendency indicates the center of the data, dispersion shows how much the values differ from each other and from the central value. Key measures of dispersion include the range, which is the difference between the highest and lowest values; the variance, which measures the average of the squared differences from the mean; and the standard deviation, which is the square root of the variance and represents the average amount by which values deviate from the mean. Another important measure is the interquartile range (IQR), which describes the spread of the middle 50% of the data. Together, central tendency and dispersion provide a more complete understanding of the data's distribution and variability.

As Dispersion measures the spread or variability of your data. A simple measure is the range, which can be calculated using `=MAX(range)-MIN(range)`, which subtracts the smallest value from the largest.

For a more statistical measure of spread, you can calculate variance. Use `=VAR.S(range)` for a sample variance or `=VAR.P(range)` for the population variance. These functions compute the average of the squared deviations from the mean.

The standard deviation is the square root of the variance and provides a more interpretable measure of spread in the same units as the data. Use `=STDEV.S(range)` for sample standard deviation or `=STDEV.P(range)` for population standard deviation.

If you need to calculate the interquartile range (IQR), which measures the middle 50% of your data and is useful for identifying outliers, you can use $=\text{QUARTILE.EXC}(\text{range},3)-\text{QUARTILE.EXC}(\text{range},1)$, which subtracts the first quartile (Q1) from the third quartile (Q3).

The central tendency and dispersion of data can be determined by Using the Data Analysis Toolpak also, many of these statistics can be computed automatically using the Data Analysis Toolpak. After activating Data Analysis Toolpak via File > Options > Add-ins, go to the Data tab and select Data Analysis. Choose Descriptive Statistics, select your data range, check the Summary Statistics box, and Excel will generate a table that includes the mean, median, standard deviation, minimum, maximum, range, and more.

Microsoft Excel - Edited_PracticalUnit1Figures

File Edit View Insert Format Tools Data Window Help

Type a question for help

D33

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	Table 3: Lives of 100 Electric Bulbs						Class Boundaries (Bins)		Frequency (f)	Average Life In Hours (x)	$y = (x - 950.5) / 80$	$f \cdot y$	$f \cdot y^2$							
1																				
2	511.6	977.7	600.2	1099.7	803.7		430.55	510.55	0	470.55	-6	0	0		CALCULATIONS USING FORMULAS					
3	923.4	1108.3	906.7	759.6	1111.9		510.55	590.55	1	550.55	-5	-5	25							
4	918.3	1051.1	992.5	817.2	665.3		590.55	670.55	2	630.55	-4	-8	32	Based on Raw Data	MEAN	978.26				
5	1143.6	848.4	939.8	1163.0	715.2		670.55	750.55	5	710.55	-3	-15	45		SD	157.63				
6	936.1	750.5	991.2	1199.5	950.2		750.55	830.55	8	790.55	-2	-16	32		MEDIAN	972.10				
7	1161.7	1027.7	995.1	966.5	1146.5		830.55	910.55	13	870.55	-1	-13	13							
8	848.0	956.8	1100.0	955.2	1023.0		910.55	990.55	26	950.55	0	0	0							
9	900.5	982.3	699.2	1069.8	1245.3		990.55	1070.55	20	1030.55	1	20	20							
10	1059.5	1091.0	850.7	1219.3	1012.6		1070.6	1150.55	12	1110.55	2	24	48							
11	1053.2	939.5	777.8	749.6	980.8		1150.6	1230.55	6	1190.55	3	18	54							
12	1091.3	1016.3	930.4	1242.2	1131.4		1230.6	1310.55	6	1270.55	4	24	96							
13	1314.7	1137.2	763.1	1294.4	917.3		1310.6	1390.55	1	1350.55	5	5	25							
14	1204.1	980.1	922.3	1057.7	907.2		More		0	1430.55	6	0	0							
15	808.0	857.7	1127.1	934.3	1262.3				100			34	390							
16	965.4	873.4	955.1	806.5	1033.0				$\mu' =$		0.34									
17	1068.3	950.3	930.6	1000.1	898.5				Mean $\mu =$		977.75									
18	1293.1	940.9	1293.8	1035.2	706.0				Variance $\sigma^2 =$		24220.16									
19	880.9	912.2	803.5	922.6	846.1				Standard Deviation $\sigma =$		155.63									
20	1092.3	1182.0	985.2	945.3	835.0															
21	1001.5	1048.8	895.1	1067.2	1062.8															
22																				
23																				
24																				
25																				
26																				
27																				

Figure1 / Figure2 / Figure3 / Figure4 / Figure5 / Figure7 / Figure8 / Figure9 / Figure10 / Figure11 / Figure12 / Figure13 / Figure14

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start

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Figure 9.3: Applying the measures of Central tendency and dispersion

Let us discuss them with the help of the Example-1 of section 9.4 above. The following figure shows an implementation.

Please note that we have used the following procedure in the worksheet to arrive at the result shown in Figure 3:

- In the cell I2 to I14, you can use the array formula as explained in the last section to calculate the frequencies of each class.
- For calculating the Average life in hours (x_j) or the class mid-points, you may enter in the cell J2 the formula $=(G2+H2)/2$.
- Copy this formula from cell J3 to J13. Notice that you need to find the value for cell J14 using different formula.
- Calculate the value of y_j in the cell K2 as per the formula $=(J2-\$J\$8)/80$. Copy this formula to cells K3 to K14.

- Now enter the formula for calculating $y_j f_j$ and $y_j^2 f_j$ (you can do it yourself).
- Enter the formula for the μ' (see page 30, Unit 1) - the mean for x -observation, variance for x -observations and Standard deviation for x -observation.

You can compare the results so obtained with the results that you can obtain by directly applying the formula of mean (AVERAGE), median (MEDIAN) and standard deviation (STDEV) on the cell range A2:E21 containing the original data.

☞ Check Your Progress 4

1. Construct a frequency distribution for the data given below, using spreadsheet package.

Table 9.2: Bike Running Data (Km) per Month

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

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2. Create the histogram from this data using Spreadsheet package.
.....
.....
.....
3. Create the measures of central tendencies and measure of dispersion for this data.
.....
.....
.....
4. What are the three main measures of central tendency?
.....
.....
.....
5. How can you calculate standard deviation in Excel?
.....
.....
.....
6. What does the interquartile range (IQR) measure?
.....
.....
.....
7. What Excel tool provides multiple descriptive statistics in one output?
.....
.....
.....

9.6 DESCRIPTIVE DATA ANALYSIS AND ITS INTERPRETATION

Descriptive data analysis is the foundational step in understanding the characteristics and structure of a dataset. It involves summarizing and organizing data using statistical measures such as mean, median, mode, standard deviation, and frequency distributions. Through tables, charts, and graphical representations, this analysis provides an overview of the data's central tendencies, dispersion, and patterns. The interpretation of these descriptive statistics is crucial, as it offers insights into trends, anomalies, and relationships within the data, guiding further analysis and decision-making processes.

explain with an example to use excel for descriptive data analysis also interpret the results obtained after descriptive data analysis is done in excel.

Since Computer is a discipline, that facilitates the working of our day-to-day life, we choose an example/Case from our daily life in this context and use it to interpret the respective results. Through the case study. This case will help us to understand, how to use Descriptive statistics as a tool for data analysis.

CASE DESCRIPTION: As a pilot project, a private company involved in hardware and maintenance related activity planned to launch its own brand of assembled computers. The company identified a small colony having 12 families in all. Each of the family is having members of different age group. For the sake of their brand performance related study, they decided to issue one computer system to each family as a single unit. They got the system installed with several applications suitable for each age group viz., Computer Games, Movies, Application Softwares, E-Books, Internet, and Security related software. The company wants to study the application usage pattern and the satisfaction level of the families, so that they can customize their product accordingly. In order to monitor the system usage time, they installed an automatic clock with the system, which records the usage time. This will help them to analyze the questionnaire feedback, which they wish to get filled, at the end of their study.

On the basis of an appraisal of the case cited above, you may identify the objectives of the study to be as under:

- What is the customer satisfaction level?
- Which software utility is performed exhaustively in most of the families?

In view of these objectives, you are required to draft a Questionnaire, where the responses to the questions are either direct or indirect, as desired. They can be categorized into one among the following scales, viz.,

- Nominal
- Ordinal
- Quantitative.

A sample questionnaire is given below, this will let you understand the Excel operations, to be performed at the later stages

QUESTIONNAIRE

Family No.:

What is the Average System Utilization time(minutes) ? :

15 30 45 60 75 90 105 120 135 150 165 180

What are the Activities Performed by your family while utilizing the system?

- a) Computer Games, b) Internet, c) E-Books,
d) Application Software's, e) Movies f) Security related software

What is the level of satisfaction, achieved by your family ?

-2 = Very dissatisfied, 1 = dissatisfied, 0 = indifference,

1 = satisfied, 2 = Very satisfied.

Would your family like to purchase the product?

Yes

No

We got the questionnaire filled from respective families, finally we have the following data from the computer savvy families. The collected data is through following variables

- *Family* (Nominal scale) is the observation/serial number of the questionnaire. In the present case, one questionnaire per household is filled by each family.
- *Time* (Quantitative scale) is a quantitative data type which is measured in minutes. It is the software utilization time for a family.
- *System Utilization Activity* (Nominal scale) is a nominal data type consisting of 6 choices of activity in the park:
 1. Computer Games,
 2. Internet,
 3. E-Books,
 4. Application Software's,
 5. Movies
 6. Security related software

To this question, there may be multiple choices that for each family, one corresponding to each of the several activities they perform.

- *Product Satisfaction* (Ordinal scale) It measures family satisfaction toward the Product. It is measured in the ordinal scale with 5 values as options listed below:
 - 2 = Very dissatisfied,
 - 1 = dissatisfied,
 - 0 = indifference,
 - 1 = satisfied,
 - 2 = Very satisfied.
- *Product Purchase* (Nominal scale) is measured in the nominal scale (Yes or No) regarding the final decision about purchase of the product.

Based on the responses to the questions framed in the questionnaire, collected data is presented in the following table:

Raw Data

Family	System Utilization Time	System Utilization Activity	Product Satisfaction	Product Purchase
1	30	1, 2, 3	0	N
2	30	4,6	1	Y
3	60	1, 2	2	Y
4	45	5	-1	N
5	30	6	1	N
6	60	2	2	Y
7	30	4	1	N
8	45	3, 4	-1	N
9	15	6	1	Y
10	60	2	2	Y
11	180	1, 2, 3, 4	2	Y
12	120	1,2,4	2	Y

Raw data presented in the table above is coded into the following form, which is amenable for further statistical analysis using spreadsheet software.

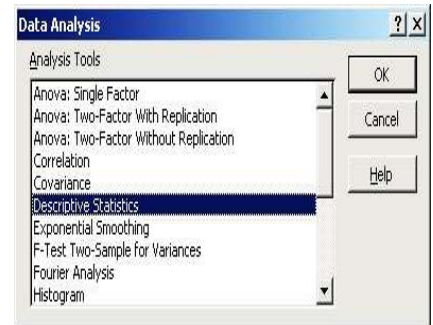
Raw Data coded for analysis in Formatted Excel Sheet

Family	System Utilization Time	System Utilization Activity						Product Satisfaction	Product Purchase
		1	2	3	4	5	6		
1	30	1	1	1	0	0	0	0	0
2	30	0	0	0	1	0	1	1	1
3	60	1	1	0	0	0	0	2	1
4	45	0	0	0	0	1	0	-1	0
5	30	0	0	0	0	0	1	1	0
6	60	0	1	0	0	0	0	2	1
7	30	0	0	0	1	0	0	1	0
8	45	0	0	1	1	0	0	-1	0
9	15	0	0	0	0	0	1	1	1
10	60	0	1	0	0	0	0	2	1
11	180	1	1	1	1	0	0	2	1
12	120	1	1	0	1	0	0	2	1

After the excel file with data in the format shown above has been created from the Raw Data table, notice that the responses under the head “*Product Purchase*” are marked 0 and 1, which represents the codes for responses “Y”

and “N”, respectively. The same also applies to the responses under “*System Utilization Activity*” head of Raw Data table.

Please note that Family, System Utilization Activity, Product Purchase, are variables measured in NOMINAL SCALE. So, we should NOT calculate sum, average, variance etc. Instead the best is to calculate frequency, percentage, plot graphs only. But here we are going to demonstrate that how you can use the tool of data analysis tool pack to get the desired answers, related to the marked objectives of your study. Now to get the answer for the stated objectives, we use the Descriptive Statistics Tool of Data Analysis ToolPak.



Enable the Data Analysis ToolPak:

- Go to File > Options > Add-ins.
- In the Manage box, select Excel Add-ins and click Go.
- Check the Analysis ToolPak box and click OK.

Data analysis through Excel:

Perform Following Steps:

1. Tabulate the collected data as given above in Excel Spreadsheet.
2. Click DATA TAB → DATA ANALYSIS → DESCRIPTIVE STATISTICS → OK

We present below some of the relevant screenshots in the present case.

Results of DESCRIPTIVE STATISTICS Procedure

Questionnaire Excel.xls [Compatibility Mode] - Microsoft Excel

FileHomeInsertPage LayoutFormulasDataReviewView

CutCopyFormat PainterClipboard

Font

Align10A⁺A⁻

B**I**U

Wrap Text

General

\$ % + < > .00 < >

Conditional Formatting as Table Styles

Format as Table Styles

Cell Styles

InsertDelete Cells

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Formatted Data										
	Family	System Utilization Time	System Utilization Activity						Product Satisfaction	Product Purchase
			1	2	3	4	5	6		
1	1	30	1	1	1	0	0	0	0	0
2	2	30	0	0	0	1	0	1	1	1
3	3	60	1	1	0	0	0	0	2	1
4	4	45	0	0	0	0	1	0	-1	0
5	5	30	0	0	0	0	0	1	1	0
6	6	60	0	1	0	0	0	0	2	1
7	7	30	0	0	0	1	0	0	1	0
8	8	45	0	0	1	1	0	0	-1	0
9	9	15	0	0	0	0	0	1	1	1
10	10	60	0	1	0	0	0	0	2	1
11	11	180	1	1	1	1	0	0	2	1
12	12	120	1	1	0	1	0	0	2	1

DESCRIPTIVE STATISTICS									
	System Utilization Time	1	2	3	4	5	6	Product Satisfaction	Product Purchase
Mean	58.75	0.33333	0.5	0.25	0.417	0.08	0.25	1	0.58333333
Standard Error	13.4970956	0.1420	0.05	0.12	0.149	0.08	0.121	0.326659474	0.1486471
Median	45	0	0.5	0	0	0	0	1	1
Mode	30	0	1	0	0	0	0	2	1
Standard Deviation	46.76464282	0.49237	0.52	0.45	0.58	0.29	0.452	1.0289295	0.51932865
Sample Variance	2186.83888	0.24242	0.27	0.2	0.265	0.08	0.205	1.272727273	0.2659552
Kurtosis	3.764955106	-1.65	-2.44	-0.33	-2.26	12	-0.33	-0.306734694	-2.2629571
Skewness	1.95028101	0.8124	0	1.33	0.388	3.46	1.327	-0.98773125	-0.3884033
Range	95	1	1	1	1	1	1	3	1
Minimum	15	0	0	0	0	0	0	-1	0
Maximum	180	1	1	1	1	1	1	2	1
Sum	705	4	6	3	5	1	3	12	7
Count	12	12	12	12	12	12	12	12	12
Confidence Level(95.0%)	29.7103847	0.31283	0.33	0.29	0.327	0.18	0.297	0.796795678	0.3277006

SAMPLE SIZE :	46.1	46.1	46	46	46.1	46	46.1	46.1	46.1
---------------	------	------	----	----	------	----	------	------	------

COLLECTED DATA				
DATA TYPE	QUANTITATIVE	ORDINAL	NOMINAL	ORDINAL
Family	System Utilization Time	System Utilization Activity	Product Satisfaction	Product Purchase
1	30	1, 2, 3	0	N
2	30	4, 6	1	Y
3	60	1, 2	2	Y
4	45	5	-1	N
5	30	6	1	N
6	60	2	2	Y
7	30	4	1	N
8	45	3, 4	-1	N
9	15	6	1	Y
10	60	2	2	Y
11	180	1, 2, 3, 4	2	Y
12	120	1, 2, 4	2	Y

Sample size = $(Z \cdot \text{Std Dev.} / \text{Std Error})^2$:
z=1.96 @ 95%CL & 1.65 @ 90% CL

Statistics	System Utilization Time	System Utilization Activity						Product Satisfaction	Product Purchase
		1	2	3	4	5	6		
Mean	58.75	0.33	0.50	0.25	0.42	0.08	0.25	1	0.58
Standard Error	13.50	0.14	0.15	0.13	0.15	0.08	0.13	0.33	0.15
Median	45	0	0.5	0	0	0	0	1	1
Mode	30	0	1	0	0	0	0	2	1
Standard Deviation	46.76	0.49	0.52	0.45	0.51	0.29	0.45	1.13	0.51
Sample Variance	2186.93	0.24	0.27	0.20	0.27	0.08	0.20	1.27	0.27
Kurtosis	3.8	-1.65	-2.44	-0.33	-2.26	12	-0.33	-0.34	-2.26
Skewness	2.0	0.81	0	1.33	0.39	3.46	1.33	-0.91	-0.39
Range	165	1	1	1	1	1	1	3	1
Minimum	15	0	0	0	0	0	0	-1	0
Maximum	180	1	1	1	1	1	1	2	1
Sum	705	4	6	3	5	1	3	12	7
Count	12	12	12	12	12	12	12	12	12
Sample size = (Z*Std Dev./Std Error) ²	46.0992	46.099	46.099	46.099	46.099	46.099	46.099	46.099	46.0992
Z=1.96 @ 95%CL & 1.65 @ 90% CL						46.099	46.099	46.0992	46.0992

DATA INTERPRETATION

Now, let us interpret the results of descriptive statistics and analyse, what they say about our objectives viz.

- What is the customer satisfaction level?
- Which software utility is performed exhaustively in most of the families?

So far as *customer satisfaction level* is concerned, the tabulated data under head of *Mean* and *Mode* gives the result as 1 and 2 respectively i.e., on an average the customers are satisfied and in fact they are “most frequently” - very satisfied. To interpret, we are to refer our data coding standards. As in this case they are :

- *Product Satisfaction* is an ordinal scale with 5 values:
-2 = Very dissatisfied,

- 1 = dissatisfied,
- 0 = indifference,
- 1 = satisfied,
- 2 = Very satisfied.

Similarly, in order to study the objective “*Which software utility is performed exhaustively in most of the families? Refer to the results under the head SUM for the **System Utilization Activity**. Here, the activity 2 is identified to be performed the most i.e. Families are mostly using the Computer system for internet related activities.*

To interpret, we are to refer our data coding standards. As in this case, they are:

- *System Utilization Activity* is a nominal data type consist of 6 choices of activity in the park:
 1. Computer Games,
 2. Internet,
 3. E-Books,
 4. Application Software’s,
 5. Movies
 6. Security related software

Now, you might be wondering that the purpose of this session is to talk about sampling and we are discussing about the descriptive statistics. This is intentional, we want you to realize that sampling is not simply a formula but it is related to various factors, as in this case the number of families are limited in the target area i.e. target population is limited. By using the formula for Sample size calculation i.e. $Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$; we can calculate the respective sample size as

$$Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$$

You might have located this expression $Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$, its just another formula for calculating the sample size, when standard deviation is known. However when standard deviation is not known, we use another formula, we are not going to discuss that, here we are going to concentrate on $Sample\ size = (Z * Std\ Dev. / Std\ Error)^2$

We calculated the sample size to be 46 families but our study is considering only 12. So for the reliability of the study the collected data is insufficient and thus we cannot conclude that the interpretation of the results related to our objectives is fine enough.

You might be in the position to realize the importance of sampling now, because from the above case, you identified that by limited data you might find that the objectives of the study are fulfilled but which might not be the fact as the sample size is inappropriate.

9.7 SUMMARY

This unit titled "Excel for Descriptive Data Analysis" provides a comprehensive guide on how to utilize Microsoft Excel for performing descriptive statistical analysis. It begins by explaining the importance of descriptive statistics in data analysis, emphasizing how Excel can be used to import, clean, and handle datasets. It covers the processes of importing data from various formats like CSV, XML, Access, and web sources. The document elaborates on data handling and pre-processing techniques such as sorting, filtering, dealing with missing values, and transforming data for analysis. It highlights frequently used Excel functions, formulas, and commands that assist in organizing and manipulating data efficiently.

Further, the unit delves into key statistical concepts such as frequency distribution, central tendency, and dispersion, explaining how to compute and interpret these using Excel. A practical case study is provided to demonstrate descriptive data analysis through a real-world example involving software usage and customer satisfaction. The study uses Excel's built-in functions and the Data Analysis Toolpak to calculate statistical measures like mean, standard deviation, and frequency, along with visualizations like histograms. The unit concludes by stressing the importance of sample size in making reliable statistical inferences, thereby linking descriptive analysis with broader statistical practices like sampling and inference.

9.8 ANSWERS TO CHECK YOUR PROGRESS

☞ Check Your Progress 1

1. What file types can Excel import data from?

A1: CSV, TXT, XML, JSON, Access, and web pages.

2. What is the importance of maintaining data consistency during import?

A2: To prevent data distortion and ensure accurate analysis.

3. What steps are followed to import data from a CSV file?

A3: Go to Data tab → From Text/CSV → Select file → Import → Load.

4. How can data be imported from a web page into Excel?

A4: Use Data → Get Data → From Web → Enter URL → Select data → Load.

☞ Check Your Progress 2

1. How do data handling and pre-processing differ?

A1: Data handling focuses on structuring data; pre-processing ensures data quality.

2. What are common Excel functions for handling text data?

A2: TRIM, LEFT, RIGHT, MID, and PROPER.

3. What is the role of commands like 'Remove Duplicates' and 'Sort'?

A3: They clean and organize data directly through Excel's interface without formulas.

4. What Excel features help with missing or incorrect data?

A4: Functions like IFERROR, ISBLANK, and Data Validation help clean such issues.

☞ **Check Your Progress 3**

1. What is the purpose of a frequency distribution?

A1: To show how often different values or ranges occur in a dataset.

2. How is the FREQUENCY function used in Excel?

A2: As an array formula: =FREQUENCY(data_range, bins_range), using CTRL+SHIFT+ENTER.

3. What tool is used in Excel to create histograms?

A3: The Data Analysis Toolpak's Histogram option.

4. How do you determine class intervals for a frequency distribution?

A4: Use the dataset's minimum and maximum values and decide on equal-width bins.

☞ **Check Your Progress 4**

1. Construct a frequency distribution for the data given below, using spreadsheet package.

Table 2: Bike Running Data (Km) per Month

511.6	977.7	600.2	1099.7	803.7
923.4	1108.3	906.7	759.6	1111.9
918.3	1051.1	992.5	817.2	665.3
1143.6	848.4	939.8	1163.0	715.2
936.1	750.5	991.2	1199.5	950.2
1161.7	1027.7	995.1	966.5	1146.5
848.0	956.8	1100.0	955.2	1023.0
900.5	982.3	699.2	1069.8	1245.3
1059.5	1091.0	850.7	1219.3	1012.6
1053.2	939.5	777.8	749.6	980.8
1091.3	1016.3	930.4	1242.2	1131.4
1314.7	1137.2	763.1	1294.4	917.3
1204.1	980.1	922.3	1057.7	907.2
808.0	857.7	1127.1	934.3	1262.3
965.4	873.4	955.1	806.5	1033.0
1068.3	950.3	930.6	1000.1	898.5
1293.1	940.9	1293.8	1035.2	706.0
880.9	912.2	803.5	922.6	846.1
1092.3	1182.0	985.2	945.3	835.0
1001.5	1048.8	895.1	1067.2	1062.8

A1. Refer section 9.5

2. Create the histogram from this data using Spreadsheet package.

A2. Refer section 9.5

3. Create the measures of central tendencies and measure of dispersion for this data.

A3. Refer section 9.5

4. What are the three main measures of central tendency?

A4: Mean, median, and mode.

5. How can you calculate standard deviation in Excel?

A5: Use =STDEV.S(range) for sample data or =STDEV.P(range) for population data.

6. What does the interquartile range (IQR) measure?

A6: The spread of the middle 50% of data ($Q3 - Q1$).

7. What Excel tool provides multiple descriptive statistics in one output?

A7: The Descriptive Statistics option in the Data Analysis Toolpak.



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